

GTIIT Spring 2025 Calculus 1M

Workshop 5 - Tuesday

March 31, 2025

Try to solve the following exercises. Feel free to chat about them with your classmates, but make sure to write your own solutions. If you get stuck, just move on to the next one and come back later.

If you need any help at all —whether it's a tiny hint, a big clue— don't hesitate to ask me!

Important note: If you find any errors or typos, please let me know so I can correct them.

1. If $f(x) = 3x^2 - x^3$, find $f'(1)$ and use it to find an equation of the tangent line to the curve $y = 3x^2 - x^3$ at the point $(1, 2)$.
2. Find $f'(a)$
 - (a) $f(x) = 3x^2 - 4x + 1$
 - (b) $f(x) = \frac{4}{\sqrt{1-x}}$
3. Give an example of a function that is continuous at $x = 0$ but not differentiable at $x = 0$.
4. If $f(x) = 2x^2 - x^3$, find $f'(x)$, $f''(x)$, $f'''(x)$ and $f^{(4)}(x)$. Graph those functions in geogebra.
5. Prove the following statements:
 - (a) The derivative of an even function is an odd function.
 - (b) The derivative of an odd function is an even function.
 - (c) Give some examples of the previous statements.
6. Use mathematical induction (p. 72 Stewart's book) to show that if $f(x) = xe^x$, then $f^{(n)}(x) = (x+n)e^x$.
7. Suppose that:
$$f(1) = 2, \quad f'(1) = 3, \quad f(2) = 3, \quad f'(2) = 2, \quad g(1) = 3, \quad g'(1) = 1, \quad g(2) = 1, \quad g'(2) = 4,$$
 - (a) If $S(x) = f(x) + g(x)$, find $S'(1)$.
 - (b) If $P(x) = f(x)g(x)$, find $P'(2)$.
 - (c) If $Q(x) = \frac{f(x)}{g(x)}$, find $Q'(1)$.
8. Let $p(x) = x^3 - x + 1$ find a polynomial $q(x)$ such that $q'(x) = p(x)$. Conclude that there exists infinitely many polynomials $q(x)$ such that $q'(x) = p(x)$.